

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 4

Status: active post-BA0-R execution plan.

Supersedes: Revision 3 only for forward execution state; R1-R3 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- No Chapter 2-4 reopen trigger is introduced by BA0-T1/T2.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. Scope correction carried into R4

DDTA evaluation begins from governed documentation intentionally authored to satisfy an analysis-readiness / portability contract.

Reliable automatic transformation of arbitrary narrative, legacy or unstructured documentation is **not** a BA0 requirement and is outside the primary thesis evaluation. Future NLP/LLM work may assist migration or candidate extraction but must retain candidate/review/provenance boundaries.

3. Base Analysis responsibility now under test

Base Analysis is the accepted methodology-neutral analytical boundary that should support, from the same governed project meaning:

- progressive human comprehension and drill-down;
- shared identity/provenance across analyses;
- method-specific projections without core contamination;
- localized diagnostics for missing/ambiguous/inconsistent information;
- governed corrective feedback;
- change-aware revalidation and targeted re-analysis.

Semantic canonicity does not imply one permanently materialized graph.

4. BA0 trial sequence

BA0-T1 - FR-3.4 application trial

Status: COMPLETED / PROVISIONAL.

Supported a shared semantic contract and origin distinctions; falsified **every useful semantic responsibility must be a first-class type**; did not prove one persisted canonical graph necessary.

Its historical successor pointer to a conflict/addition stress test is superseded by the dedicated successor-supersession note in **methodology**/.

BA0-T2 - analysis consumption, teachability and governed corrective feedback

Status: COMPLETED / PROVISIONAL PASS.

Required checks:

1. a human can navigate a progressive overview-to-source projection over the facial-access branch;
2. a concrete bounded analysis consumer can reuse the same accepted project semantics without redefining them;
3. insufficient/missing information can be returned as a source-targeted corrective documentation candidate;
4. the trial does not rely on automatic arbitrary-narrative extraction or one persisted graph.

The bounded consumer is a pre-overlay STRIDE transfer probe used only to pressure-test the shared boundary. It is not the formal O2 STRIDE implementation.

BA0-T3 - responsibility and non-goals closure review

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

Review BA0-R + T1 + T2 together and decide only:

- precise Base Analysis responsibility statement;
- explicit non-goals;
- whether progressive teachability is a required Base Analysis capability;
- whether accepted semantic identity may be reused or reproducibly rebuilt;
- whether diagnostics and governed corrective feedback belong inside the BA0 responsibility boundary;
- whether remaining uncertainty belongs to BA1-BA6 rather than BA0.

Exit: BA0 becomes CLOSED only if these responsibilities are precise enough to begin ontology work without smuggling in a pre-approved type list.

5. Sequence after BA0 closure

Do not skip phases:

BA1 - Minimal BAE ontology

BA2 - Relations and canonical action vocabulary

BA3 - Document-to-BAE derivation, provenance and authority

BA4 - Multi-level human and method projections

BA5 - Controlled vocabulary and optional authoring/extraction assistance

BA6 - Base Analysis closure and regression

BA1 - Minimal BAE ontology

Start from responsibilities forced jointly by governed DDTA corpora, BA0-R evidence and BA0 trials. No type from historical **primary-analysis-focus** studies, STRIDE, SysML, ThreatForge or another notation is accepted by default.

BA2 - Relations and canonical action vocabulary

Define the smallest stable semantic relation vocabulary. Keep semantic relation, canonical lexical label and authoring synonym assistance separate.

BA3 - Document-to-BAE derivation, provenance and authority

Close identity, source locators, grounded/derived/reviewed-addition boundaries, acceptance/rejection, stale state and document-to-analysis traceability. Automatic candidate extraction may be evaluated as optional assistance but is not required for thesis validity.

BA4 - Multi-level human and method projections

Views are derived from the same accepted Base Analysis semantics, whether the representation is reused or reproducibly rebuilt. Test both progressive human teachability and analysis-specific views. No view is source authority.

BA5 - Controlled vocabulary and optional assistance

Vocabulary reuse, duplicate suggestions, LLM/NLP proposals and candidate extraction are optional support. None may silently establish identity, equivalence, ownership, project fact or BAE acceptance.

BA6 - Base Analysis closure and regression

Regress closed corpora and at least one structurally different holdout. Verify semantic identity stability, source traceability, projection consistency, diagnostic behavior, change/re-analysis scope, method neutrality and human drill-down support.

6. Analysis envelope after Base Analysis closure

Only after the Base Analysis gate:

- A1 - AnalysisRecord / execution envelope.
- A2 - common reviewed Finding boundary.
- A3 - change/provenance integration.

Then define the generic overlay contract and full methodology evaluations.

7. Method-specific evaluations

Planned concrete evaluations remain STRIDE and STRIDE-AI over the same common core. They are demonstrators of the overlay boundary, not proof of universal method support.

The BA0-T2 STRIDE probe does not count as the formal STRIDE evaluation; it exists only to falsify the Base Analysis responsibility before ontology design.

8. Corrective feedback boundary

The general DDTA loop is broader than Finding -> SecurityRequirement:

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analysis / validation / diagnostic
-> source-localized corrective candidate
-> governed review
-> correction / clarification / supersession / new requirement or decision
-> Base Analysis revalidation
-> impacted re-analysis
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The thesis may still evaluate the narrower accepted security-Finding-to-SecurityRequirement path as a specific research contribution. BA0 must not assume every correction has the same document type.

9. Next authorized microstep

Execute only:

BA0-T3 - responsibility and non-goals closure review.

Do not start BA1, formal STRIDE overlay design, Common Finding schema or implementation work unless BA0 closes explicitly.